

MIP_WAVE2_FOUNDATION_PACK.md

Mainframe Intelligence Platform

Wave 2 Engineering Foundation Pack

Version: 1.0

Status: Approved

Contents

Batch 1

1. MIP_WORKSPACE_BOOTSTRAP.md

Purpose: Establish standard onboarding, development environment setup, repository initialization, Copilot configuration, quality gates, and engineering workflow.

Key Sections:

- Workspace Setup
 - VS Code Configuration
 - Python Environment
 - Repository Layout
 - Skills Installation
 - Prompt Library Installation
 - Development Workflow
 - Quality Gates
 - Definition of Ready
 - Definition of Done
-

2. MIP_REPOSITORY_STRUCTURE.md

Purpose: Define the official repository layout and ownership model.

Repository Structure:

```
mip-platform/  
  
.github/  
docs/  
skills/
```

```
prompts/  
  
src/  
tests/  
  
scripts/  
tools/  
  
examples/  
output/  
  
data/  
logs/  
  
deployment/
```

Core Principles:

- Clean Architecture
- Domain Driven Design
- Metadata First
- Modular Design
- Scalability

Ownership:

- Architecture Team
- Platform Team
- Engineering Team
- DevOps Team

Dependency Rules:

```
api  
↓  
services  
↓  
domain  
  
repository  
  
graph
```

3. MIP_GITHUB_COPILOT_STRATEGY.md

Purpose: Establish a repeatable AI-assisted engineering workflow.

Core Principle:

Source Code
↓
Metadata
↓
Relationships
↓
Knowledge Graph
↓
Reasoning
↓
Modernization Intelligence

Engineering Workflow:

Design
↓
Review
↓
Critique
↓
Improve
↓
Implement
↓
Test
↓
Refactor
↓
Document

Copilot Modes:

- Architect
- Analyst
- Engineer
- Reviewer

Skill-Based Engineering:

- repository-engineer
- mainframe-modernization-architect
- mainframe-code-analyst
- metadata-modeler
- graph-engineer
- sqlite-engineer
- test-engineer
- code-reviewer
- documentation-writer

Batch 2

Skill Framework

Skill 1

repository-engineer

Purpose

Design scalable repository structures, engineering standards, dependency management, package boundaries, and workspace organization.

Inputs

- Requirements
- Architecture
- Existing repositories

Outputs

- Repository layouts
- Dependency rules
- Ownership models

Responsibilities

- Repository Architecture
- Dependency Management
- Workspace Design
- Scalability Review

Constraints

- Clean Architecture
- SOLID
- Low Coupling

Success Criteria

New engineer onboarding $\leq$ 1 day.

Skill 2

mainframe-modernization-architect

Purpose

Create modernization roadmaps and service decomposition strategies.

Inputs

- COBOL
- JCL
- DB2
- Metadata
- Knowledge Graph

Outputs

- Capability Maps
- Modernization Roadmaps
- Service Boundaries

Responsibilities

- Legacy Assessment
- Domain Discovery
- Migration Planning

Constraints

Preserve:

- Business Rules
 - Transaction Integrity
 - Regulatory Requirements
-

Skill 3

mainframe-code-analyst

Purpose

Extract facts from mainframe artifacts.

Inputs

- COBOL
- JCL

- Copybooks
- DB2

Outputs

- Metadata
- Inventories
- Dependency Maps

Responsibilities

- Program Discovery
- Data Discovery
- Execution Discovery

Constraints

Never infer.

Always provide evidence.

Skill 4

metadata-modeler

Purpose

Create canonical metadata models.

Inputs

- Metadata
- Relationships

Outputs

- Entity Models
- Schemas
- Governance Standards

Responsibilities

- Canonical Modeling
 - Schema Evolution
 - Governance
-

Skill 5

graph-engineer

Purpose

Transform metadata into relationship intelligence using NetworkX.

Inputs

- Metadata
- Relationships

Outputs

- Graph Models
- Traversal Strategies

Responsibilities

- Node Modeling
 - Edge Modeling
 - Traversal Design
-

Skill 6

sqlite-engineer

Purpose

Design SQLite persistence layer.

Inputs

- Metadata
- Query Requirements

Outputs

- Schemas
- Indexes
- Migration Strategies

Responsibilities

- Schema Design
- Index Design
- Migration Readiness

Skill 7

test-engineer

Purpose

Ensure reliability and regression safety.

Inputs

- Designs
- Code
- Requirements

Outputs

- Unit Tests
- Integration Tests
- Coverage Reports

Responsibilities

- Test Planning
 - Test Generation
 - Coverage Analysis
-

Skill 8

code-reviewer

Purpose

Identify architecture and implementation risks.

Inputs

- Designs
- Code
- Pull Requests

Outputs

- Findings
- Risks
- Recommendations

Responsibilities

- Architecture Review
 - Code Review
 - Maintainability Review
-

Skill 9

documentation-writer

Purpose

Preserve engineering knowledge.

Inputs

- Architecture
- Code
- Reviews

Outputs

- READMEs
- Playbooks
- Architecture Guides

Responsibilities

- Architecture Documentation
 - Engineering Documentation
 - Knowledge Preservation
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Wave 2 Completion Status

Batch 1

- ✓ Workspace Bootstrap
 - ✓ Repository Structure
 - ✓ GitHub Copilot Strategy
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Batch 2

- ✓ repository-engineer

✓ mainframe-modernization-architect

✓ mainframe-code-analyst

✓ metadata-modeler

✓ graph-engineer

✓ sqlite-engineer

✓ test-engineer

✓ code-reviewer

✓ documentation-writer

Next Phase

Batch 3 – Prompt Library

Directories:

```
prompts/  
  
architecture/  
discovery/  
parsing/  
metadata/  
graph/  
testing/  
review/  
modernization/
```

Expected Deliverables:

- 25–30 reusable Copilot prompts
- Discovery prompts
- Parsing prompts
- Metadata prompts
- Graph prompts
- Testing prompts
- Review prompts
- Modernization prompts